BEACONHOUSE NATIONAL UNIVERSITY

LAB 7 | WEEK 7

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BNU ID: F2023-009

Due Date: 29 Mar, 2025

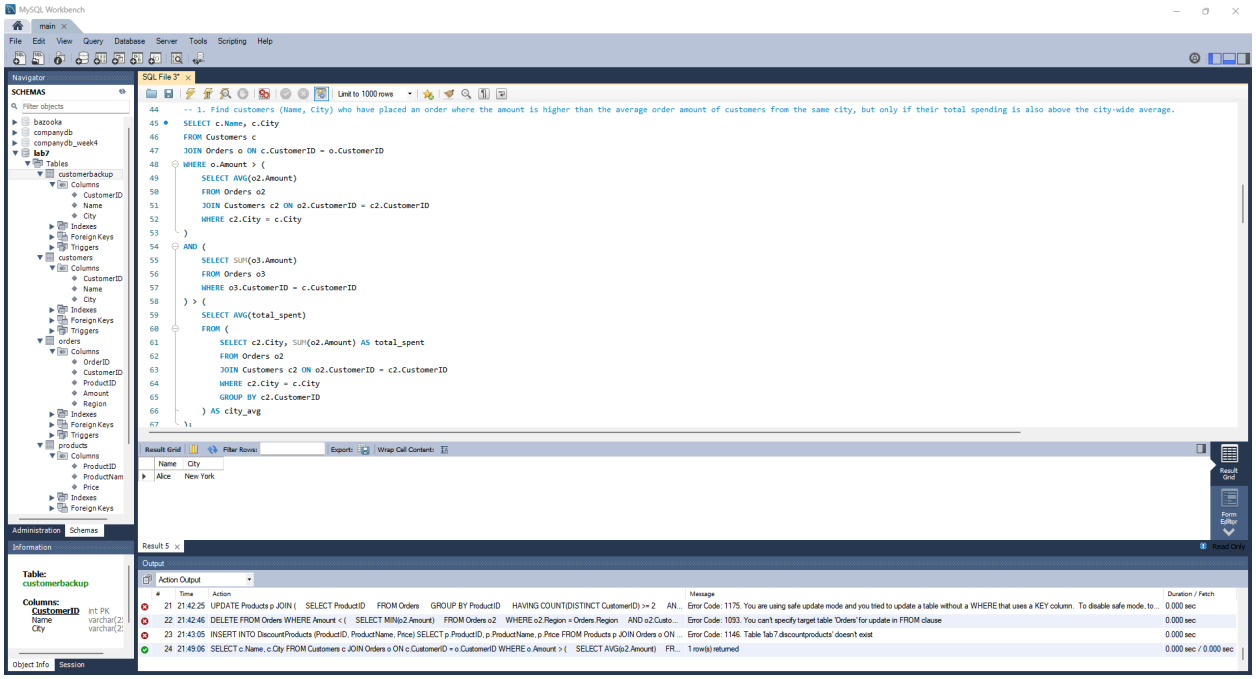
Lab Manual

For

ADMS(SEC_B)

Course Instructor	Ms. Talia Noreen
Lab Instructor	Mr. Ahsan Atiq
Semester	Spring 2025

1. Find customers (Name, City) who have placed an order where the amount is higher than the average order amount of customers from the same city, but only if their total spending is also above the city-wide average.



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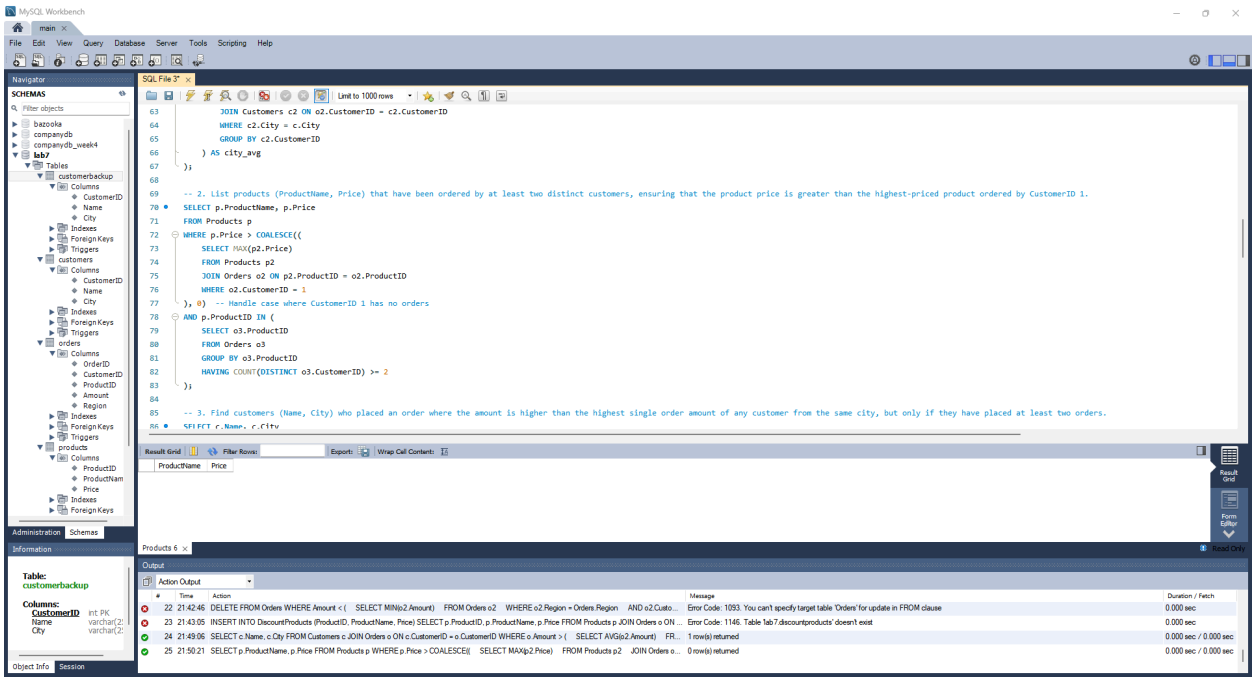
-- 1. Find customers (Name, City) who have placed an order where the amount is higher than the average order amount of customers from the same city, but only if their total spending is also above the city-wide average.
SELECT c.Name, c.City
FROM Customers c
JOIN Orders o ON c.CustomerID = o.CustomerID
WHERE o.Amount > (
    SELECT AVG(o2.Amount)
    FROM Orders o2
    JOIN Customers c2 ON o2.CustomerID = c2.CustomerID
    WHERE c2.City = c.City
)
AND (
    SELECT SUM(o3.Amount)
    FROM Orders o3
    WHERE o3.CustomerID = c.CustomerID
) > (
    SELECT AVG(total_spent)
    FROM (
        SELECT c2.City, SUM(o2.Amount) AS total_spent
        FROM Orders o2
        JOIN Customers c2 ON o2.CustomerID = c2.CustomerID
        WHERE c2.City = c.City
        GROUP BY c2.CustomerID
    ) AS city_avg
);
```

2. List products (ProductName, Price) that have been ordered by at least two distinct customers, ensuring that the product price is greater than the highest-priced product ordered by CustomerID 1.

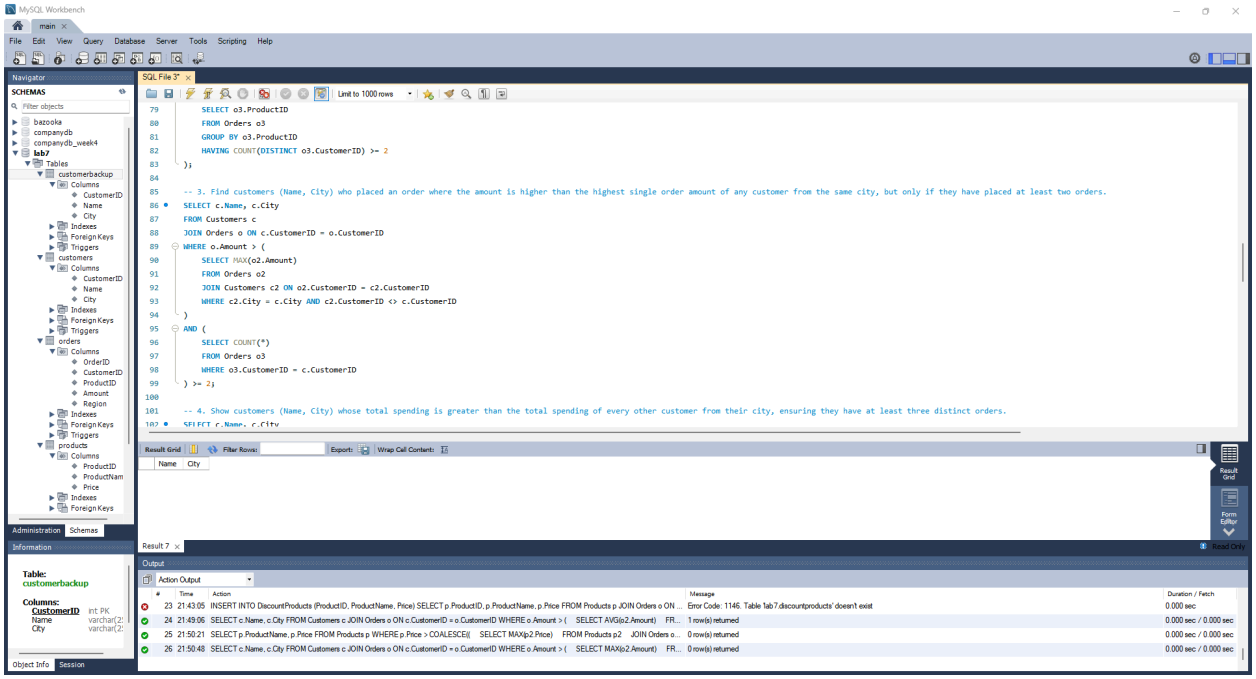


```
    JOIN Customers c2 ON o2.CustomerID = c2.CustomerID
    WHERE c2.City = c.City
    GROUP BY c2.CustomerID
) AS city_avg
);

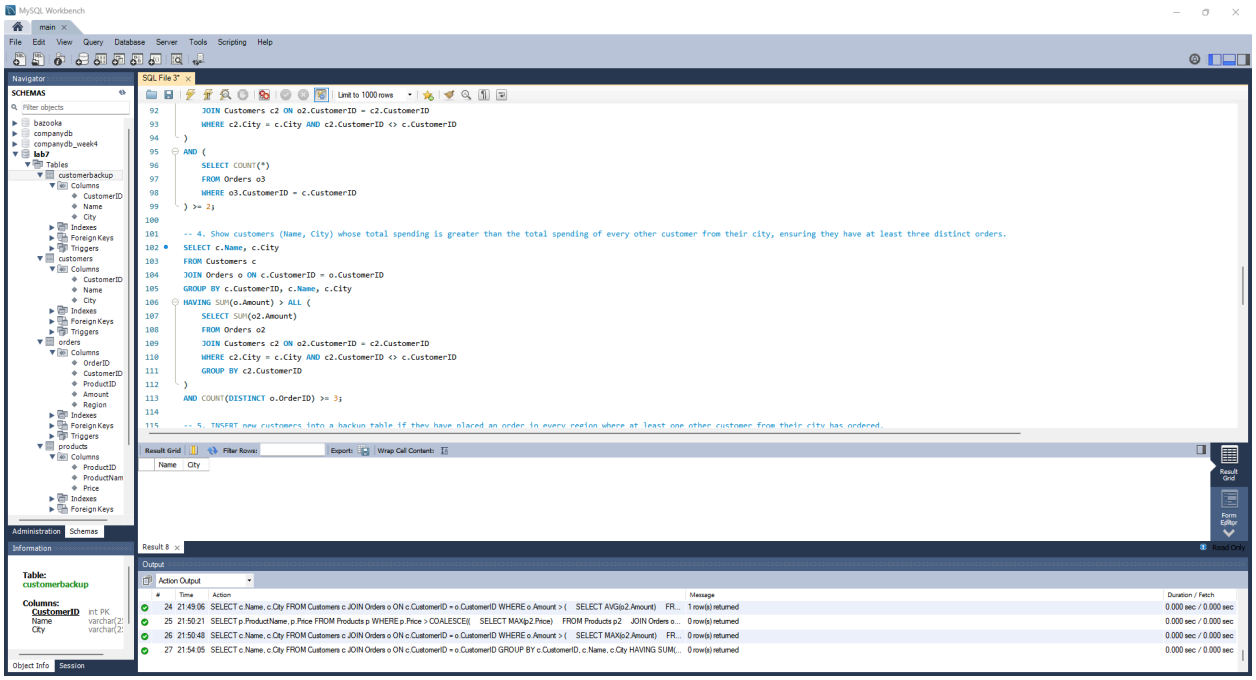
-- 2. List products (ProductName, Price) that have been ordered by at least two distinct customers, ensuring that the product price is greater than the highest-priced product ordered by CustomerID 1.
SELECT p.ProductName, p.Price
FROM Products p
WHERE p.Price > COALESCE((
    SELECT MAX(p2.Price)
    FROM Products p2
    JOIN Orders o2 ON p2.ProductID = o2.ProductID
    WHERE o2.CustomerID = 1
), 0)  -- Handle case where CustomerID 1 has no orders
AND p.ProductID IN (
    SELECT o3.ProductID
    FROM Orders o3
    GROUP BY o3.ProductID
    HAVING COUNT(DISTINCT o3.CustomerID) >= 2
);

-- 3. Find customers (Name, City) who placed an order where the amount is higher than the highest single order amount of any customer from the same city, but only if they have placed at least two orders.
SELECT c.Name, c.City
```

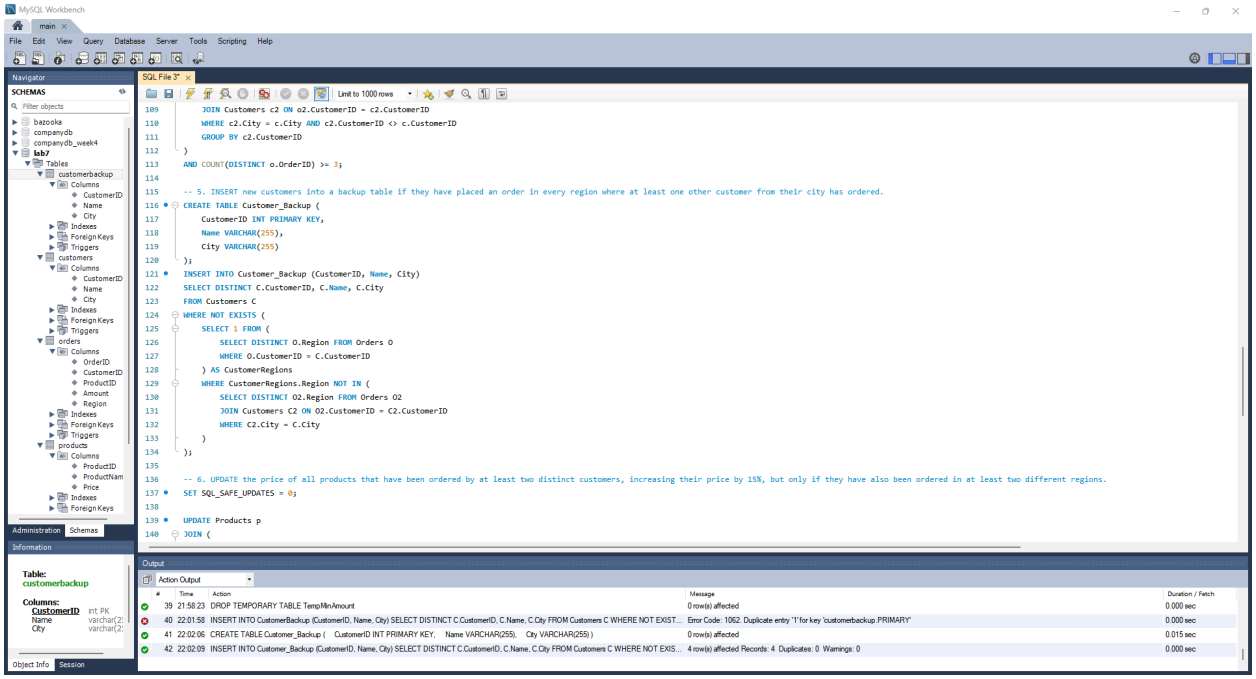
3. Find customers (Name, City) who placed an order where the amount is higher than the highest single order amount of any customer from the same city, but only if they have placed at least two orders.



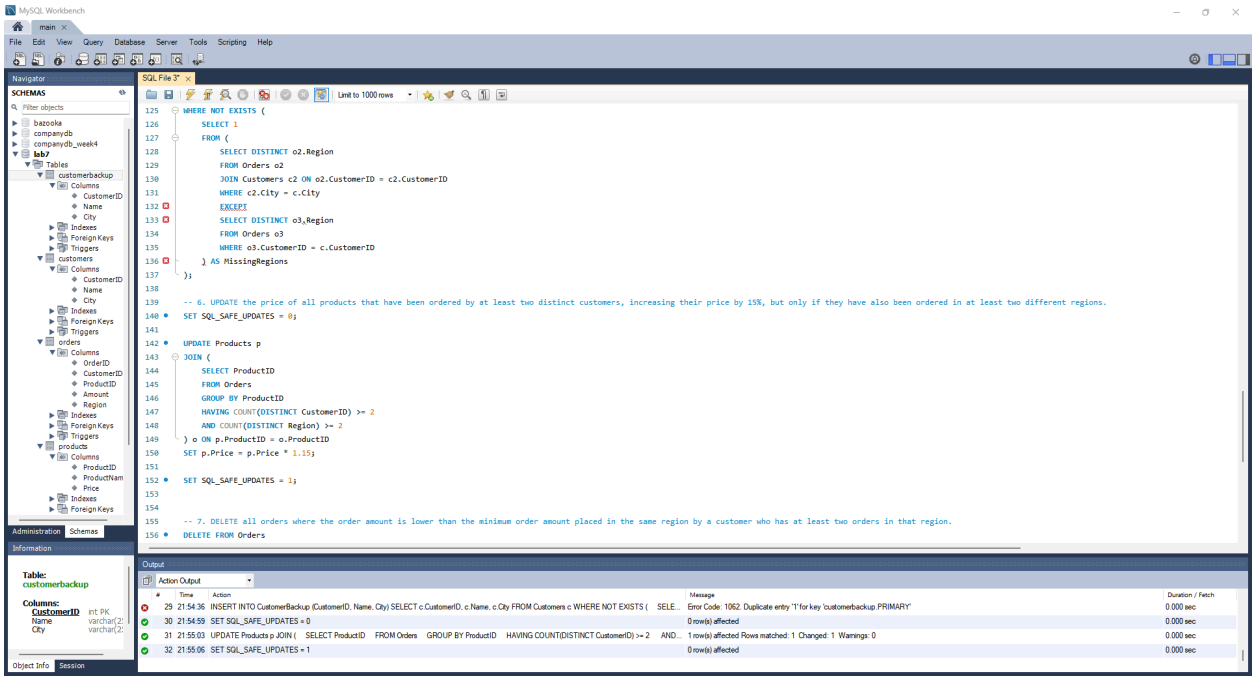
4. Show customers (Name, City) whose total spending is greater than the total spending of every other customer from their city, ensuring they have at least three distinct orders.



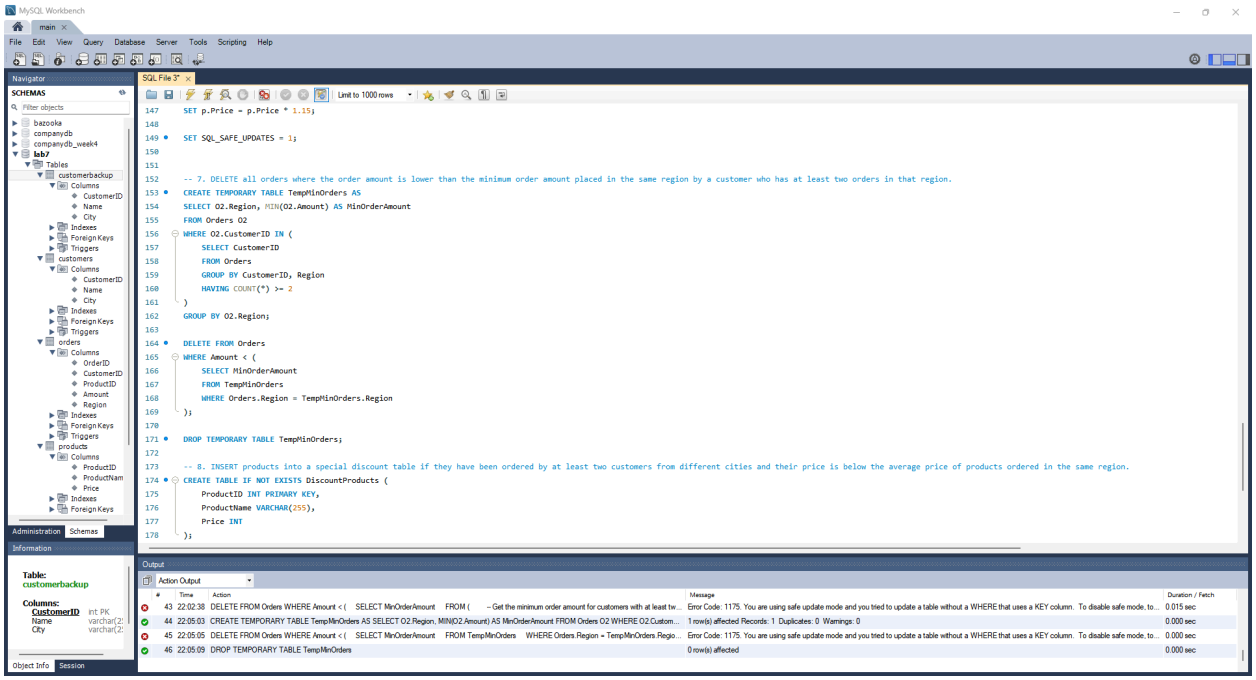
5. INSERT new customers into a backup table if they have placed an order in every region where at least one other customer from their city has ordered.



6. UPDATE the price of all products that have been ordered by at least two distinct customers, increasing their price by 15%, but only if they have also been ordered in at least two different regions.



7. DELETE all orders where the order amount is lower than the minimum order amount placed in the same region by a customer who has at least two orders in that region.



8. INSERT products into a special discount table if they have been ordered by at least two customers from different cities and their price is below the average price of products ordered in the same region.